

ANRAN (ANDY) TAN, FRM

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Education

Duke University Department of Economics 09/2024 – Present
Ph.D. in Economics

Peking University Guanghua School of Management 07/2019 – 06/2022
Master of Economics

Peking University School of Economics & School of Law 07/2015 – 06/2019
Bachelor of Economics; Bachelor of Laws (Double Degree)

Relevant Coursework: Machine Learning, Deep Learning, Financial Econometrics, Empirical Asset Pricing

Publications

“**Hierarchical Organizational Design: A Contest Design Perspective**” (in Chinese), with Xiruo Wang, Zenan Wu, Tao Hu. *China Journal of Economics*, (2025). [LINK]

Skills

Programming: Python (PyTorch, sklearn, OpenCV), R, Stata, MATLAB, Mathematica, L^AT_EX

Quantitative Methods: Financial Econometrics, High-Frequency Data Analysis, Realized Volatility Estimation (HAR-RV, Realized Kernels, Bipower Variation, TSRV), Monte Carlo Simulation, Causal Inference (DiD, RCT, IV), Time Series (ARIMA, GARCH), Machine Learning, Deep Learning, Generative Models (Diffusion, GAN), Computer Vision, NLP

Finance: Asset Pricing, Empirical Asset Pricing, Market Microstructure, Structured Finance (ABS, REITs), Financial Risk Management (**FRM** certified), Wind

Research & Projects

Evaluating Realized Estimators via Controllable Diffusion Models (Ongoing) 02/2026 – Present
Field Paper, Advisor: Prof. Andrew Patton Duke University

- Architecting a controllable generative framework using **score-based diffusion models** to synthesize tick-level order-flow data, enabling scalable **market microstructure simulation** decoupled from historical path assumptions.
- Designing a **conditional volatility injection** mechanism that parametrizes diffusion trajectories by pre-specified realized variance targets, producing synthetic price paths with controlled Itô semimartingale properties for rigorous Monte Carlo evaluation.
- Extending the robust loss-function framework of Patton (2011) to a generative-data setting; constructing comparative rankings of **realized volatility estimators** — including Realized Kernels, Bipower Variation, TSRV, and HAR-RV — across simulated liquidity regimes and market conditions.
- Evaluating estimator efficiency and asymptotic properties across distinct market regimes, quantifying estimation bias and variance under simulated extreme market conditions.
- Developing a benchmark pipeline to stress-test **high-frequency estimators** under microstructure noise, bridging generative AI with classical financial econometrics to enhance the selection of robust **volatility proxies**.

Parameter-Efficient Fine-Tuning of LLMs: LoRA / Sparse LoRA / HiRA 10/2025 – 12/2025
Course Project, Advisor: Prof. Vahid Tarokh Duke University

- Engineered a systematic benchmarking framework evaluating three PEFT methods (LoRA, Sparse LoRA, HiRA) on DistilBERT across classification (SST-2, IMDB) and language modeling (WikiText-2) tasks; automated hyperparameter sweeps via grid search.
- Conducted sensitivity analysis on rank r , scaling factor α , and batch size B ; quantified stability-efficiency trade-offs and demonstrated HiRA’s superior robustness across diverse data distributions and hyperparameter regimes.
- Formulated a model-adaptation strategy matrix linking computational budget constraints to PEFT method selection, providing actionable guidelines for LLM fine-tuning in resource-limited settings.

Naming and Pricing Strategy in Chinese Online Used Book Markets 2021
Graduation Thesis, Advisor: Prof. Xi Weng Peking University

- Constructed a high-granularity dataset from Kongfz.com via web scraping; applied KDE and nonparametric regression to identify a positive correlation between naming complexity and cross-seller price dispersion.
- Extended Crawford-Sobel (1982) cheap talk to derive equilibrium signaling strategies: high-quality sellers use non-informative keywords to screen high-value buyers; liquidity-constrained sellers adopt direct advertising to minimize search frictions.

Work Experience

Energy Policy Institute at the University of Chicago (EPIC-China)

07/2022 – 06/2024

Research Assistant

Beijing, China

- Estimated individual latent productivity via **high-dimensional fixed effects (HDFE)** on 80M unstructured verdicts; reconstructed 500K career trajectories to empirically test the Peter Principle and formulated a sequential contest model of promotion incentives.
- Constructed a 200K-profile revolving-door database; applied **Difference-in-Differences (DiD)** and Kernel Density Estimation (KDE) to quantify the causal spillover of legal expertise on firm-level litigation outcomes.
- Implemented a **computer vision pipeline** (OCR + SIFT) to extract patent and trademark data from verdicts and link with third-party databases, quantifying visual similarities to assess IP litigation risks and infringement patterns.
- Developed large-scale scrapers (Requests, Selenium, OpenCV) for government procurement and green-industry policy documents; identified regulatory shock timing via **NLP** for use in event-study identification strategies.
- Co-designed a **Randomized Controlled Trial (RCT)** on civil trial live-streaming: established experimental protocols, survey frameworks, and geospatial visualizations for environmental court coverage analysis.

ICBC UBS Asset Management

06/2020 – 10/2020

Investment Banking Intern

Beijing, China

- Developed simulation models to project **Asset-Liability** dynamic shifts for Public REITs; performed feasibility studies on IDC assets using quantitative valuation frameworks.
- Quantified investment scales and structured tranches for **Asset-Backed Securities (ABS)**; developed structured finance proposals and internal risk-mitigation frameworks.
- Engineered multi-scenario **cash flow models** for supply chain financing; synthesized regional macroeconomic indicators and real estate supply-demand data across 6 cities to enhance forecasting precision.
- Analyzed financial statements of State-Owned Enterprises (SOEs); streamlined financial modeling workflows and authored due diligence reports for non-standard asset management schemes.

Deloitte Touche Tohmatsu CPAs

01/2017 – 02/2017

Group 4 Audit Intern

Beijing, China

- Performed substantive testing and voucher verification for large-scale SOEs; reconciled primary financial records including inventory receipts and bank statements.
- Leveraged Wind and Jiuqi to consolidate financial statements; prepared consolidated audit reports under CAS. Received **EE (Exceeds Expectations)** — the highest performance grade.

Honors & Awards

Excellent Graduate (Graduated with Distinction), <i>Guanghua School of Management, Peking University</i>	06/2022
Horizon Scholarship (top 5%), <i>Peking University</i>	12/2020
College Graduate Excellence Award of Beijing City (top 2%), <i>Beijing Municipality</i>	07/2019
Excellent Graduate (Graduated with Distinction, top 5%), <i>Peking University</i>	06/2019
National Scholarship (top 2%), <i>Peking University</i>	12/2018
Hana Financial Group Scholarship, <i>Peking University</i>	12/2017

Teaching Experience

Duke University Department of Economics

Time Series Econometrics, <i>Prof. Marta Boczoń</i>	2026
Game Theory, <i>Prof. Attila Ambrus</i>	2025

University of Chicago Harris School of Public Policy

Environmental Economics and Policy Lab, <i>Prof. Shaoda Wang</i>	2023
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Peking University Guanghua School of Management

Chinese Economics, <i>Prof. Li'an Zhou, Prof. Hui Wang, and Prof. Yao Tang</i>	2022
AI and Economics, <i>Prof. Juanjuan Meng and Prof. Xi Weng</i>	2022
Topics in Industrial Economics, <i>Prof. Xing Li</i>	2022
Organizational Economics, <i>Prof. Shuo Liu</i>	2021
Advanced Microeconomics, <i>Prof. Juanjuan Meng</i>	2021
Behavioral Economics, <i>Prof. Juanjuan Meng</i>	2020

Peking University School of Economics

Principles of Economics (I), <i>Prof. Zenan Wu</i>	2022
Introduction to Chinese Economy, <i>Prof. Xi Ji</i>	2018